Use Case Specification: UC-15

- **ID / Code:** UC-15
- **Use Case Name:** Issue & Send Invoice
- **Description:** Allows an Administrator to issue an invoice in "ready to issue" state (making it legally final) and send it to the customer's billing contact, managing the invoice lifecycle status (UR-2.3, FR-2.4). Sending includes the automated notification process (UC-23).
- **Primary Actor:** Administrator
- **Scope:** Subscription Management & Invoicing System
- **Level:** User Goal
- **Stakeholders & Interests:**
  - *Administrator:* Wants to issue invoices safely, without losing or duplicating them, and to know exactly what failed if something goes wrong (UR-2.3, NFR-1.5, NFR-1.6).
  - *Customer company:* Wants to receive the invoice immediately (FR-4.1.1).
  - *Company (Product Owner):* Wants compliant, sequentially numbered invoices and payment follow-up.
- **Preconditions:** The invoice is in "ready to issue" state; the customer has an active billing contact; the period is open.
- **Postconditions:** The invoice is "issued" and, if delivery succeeds, "sent".
- **Success Guarantee:** The invoice is issued with its final number and PDF, sent to the billing contact, its state changes to "sent" (FR-2.4), administrators are notified of the issuance (FR-4.2.1) and follow-up reminders are scheduled (FR-4.1.2).
- **Minimal Guarantee:** An invoice is issued only once even if the page is reloaded (NFR-1.5); if sending fails, the invoice remains "issued" and the error is reported distinctly from a creation error (NFR-1.6).
- **Trigger:** The Administrator selects "Issue & send" on an invoice.
- **Main Success Scenario:**

1. **Administrator:** Selects an invoice and requests to issue and send it.
2. **System:** Shows a preview and warns that issuing is a legally irreversible action (NFR-1.5).
3. **Administrator:** Confirms.
4. **System:** Validates the invoice data and verifies that the financial period is open. It assigns a unique sequential invoice number (FR-2.2.2), generates the final PDF and changes the invoice state to "issued" (FR-2.4).
5. **System: Includes UC-23** to send the invoice to the customer's billing contact and to notify administrators of the issuance (FR-4.1.1, FR-4.2.1).
6. **System:** Changes the invoice state to "sent" and schedules follow-up reminders 3 and 5 calendar days after issuance (FR-4.1.2), adjusting dates that fall on non-working days (FR-4.4.2).
7. **System:** Displays the result to the Administrator; for batch operations, shows task progress asynchronously (NFR-1.4).

- **Extensions:**
  - 3a. Administrator cancels:
    - 3a1. The invoice remains "ready to issue".

- 4a. Missing data or no active billing contact:
  - 4a1. System blocks issuance and identifies the missing mandatory invoice information.
- 4b. Period locked (extension point *period locked*):
  - 4b1. System blocks issuance (FR-2.7); a Super-Administrator override (UC-17) is required.
- 5a. Delivery to the customer fails:
  - 5a1. System keeps the invoice "issued", queues a retry and shows a specific message: "Invoice was issued but could not be sent" (NFR-1.6).
  - 5a2. System records the delivery failure and makes it visible in the system health dashboard (FR-6.10).
- *a. Page reloaded during issuance:
  - *a1. System detects the operation already in progress/completed and does not issue the invoice twice (NFR-1.5).